# A COMPREHENSIVE GUIDE TO ICT THESIS WRITING FOR BACHELOR AND MASTER DEGREES

This document provides a complete, structured, and internationally accepted guide for writing Bachelor and Master theses. It integrates best practices from highly successful theses worldwide, especially in engineering, ICT, health, AI, and applied system development research.

## General Explanatory Speech for the Thesis Template

This thesis is structured in accordance with internationally recognized Anglo-Saxon academic standards, which emphasize clarity, logical progression, methodological rigor, and evidence-based reasoning. The organization of the work follows a coherent sequence that moves from problem identification to solution design, implementation, evaluation, and critical reflection.

The study begins by establishing the general context and motivation of the research, ensuring that the problem under investigation is clearly situated within its scientific, social, and practical environment. This initial framing enables the reader to understand the relevance of the topic and the necessity of the research. The problem is then articulated in precise terms, followed by the formulation of research questions and objectives that guide the entire study.

Building on this foundation, the thesis reviews existing theories, models, systems, and empirical studies related to the research domain. This review serves to demonstrate mastery of current knowledge, critically analyze prior work, and identify gaps or limitations that justify the proposed research. The literature review thus provides the theoretical and conceptual basis upon which the study is developed.

Following the identification of these gaps, the thesis presents the research methodology and the design approach adopted to address the problem. This includes the selection of an appropriate research paradigm, the definition of models or principles guiding the system, and the description of the tools, techniques, and metrics used. This stage translates theoretical considerations into a structured and reproducible design framework.

The work then proceeds to the implementation and evaluation phase, where the proposed system or solution is realized and rigorously tested. Implementation details are provided to ensure reproducibility, while evaluation results are analyzed using predefined metrics. Where applicable, comparisons with existing solutions are presented to objectively assess performance and relevance.

Finally, the thesis concludes by synthesizing the main findings, highlighting the scientific, technical, and practical contributions of the work. Limitations encountered during the study are discussed transparently, and perspectives for future research and system enhancement are proposed. This closing reflection ensures that the research is not only complete but also open to further development and real-world application.

Overall, this structured approach ensures that the thesis is logically consistent, academically sound, and aligned with the expectations of international examiners, while remaining adaptable to a wide range of disciplines and applied research contexts.

**The guide is adaptable to technical, interdisciplinary, and applied research, including system development, mobile applications, assistive technologies, and health-ICT research.**

# PRELIMINARY PAGES (MANDATORY)

Before Chapter 1, the following pages are expected in most Anglo-Saxon universities:

1. **Title Page**
    - University, Faculty, Department
    - Thesis title (clear, concise, solution-oriented)
    - Author name
    - Degree (Bachelor/Master)
    - Supervisor(s)
    - Academic year
2. **Declaration / Statement of Originality**
    - Confirms the work is original and properly cited
3. **Certification**
4. **Academic Approval Page** (if required)
5. **Dedication** (optional)
6. **Acknowledgements** (optional but recommended)
7. **Abstract** (VERY IMPORTANT)
    - 250–300 words (Master), 150–250 words (Bachelor)
    - Problem → Methodology → System/Model → Key Results → Contribution
8. **Keywords** (5–7)
9. **Table of Contents**
10. **List of Figures**
11. **List of Tables**
12. **List of Acronyms and Abbreviations**

# CHAPTER 1: GENERAL INTRODUCTION

**(Purpose of the Chapter**

To frame the research, justify its relevance, define the problem, and present the research roadmap.)

## 1.1 Context of the Study

- Global context (scientific, technological, societal)
- Regional/National context (e.g., Cameroon, Africa, LMICs)
- Sectoral context (health, ICT, disability, education, etc.)
- Why the topic is timely and relevant

## 1.2 Background of the Study

- Historical evolution of the problem
- Existing solutions and their evolution
- Key stakeholders involved
- Policy, standards, or frameworks (WHO, ISO, WCAG, etc. if applicable)

## 1.3 Problem Statement

- Clear, concise, evidence-based
- Describe the current situation, undesired condition, and consequences
- Avoid solutions here

## 1.4 Research Gap

- What existing studies do not address adequately
- Technological, methodological, contextual, or population gap
- Explicit justification of novelty

## 1.5 Research Questions

- Main Research Question
- Specific Research Questions (3–6)
- Must be answerable and measurable

## 1.6 Research Objectives

- **General Objective** (aligned with main question)
- **Specific Objectives** (aligned with specific questions)

## 1.7 Research Hypotheses (if applicable)

- Null and alternative hypotheses (mainly for quantitative studies)

## 1.8 Significance of the Study

- Scientific contribution
- Social impact
- Technical/industrial relevance
- Policy or institutional relevance

## 1.9 Scope and Delimitation of the Study

- Geographical scope
- Population scope
- Technical scope
- Time frame

## 1.10 Limitations of the Study

- Data constraints
- Technical limitations
- Environmental or ethical constraints

## 1.11 Ethical Considerations

- Consent, privacy, accessibility, inclusion
- Ethical approvals if applicable

## 1.12 Structure of the Thesis (Plan / “Advertissement”)

- Brief description of each chapter

# CHAPTER 2: LITERATURE REVIEW AND THEORETICAL STUDY

**(Purpose of the Chapter**

To demonstrate mastery of existing knowledge, position the study academically, and justify the proposed solution).

## 2.1 Introduction to the Literature Review

- Objectives of the chapter
- Review strategy (databases, keywords)

## 2.2 Definition of Key Concepts and Terms

- Operational definitions
- Technical definitions
- Regulatory definitions

## 2.3 Theoretical and Conceptual Framework

- Fundamental principles
- Models, theories, or frameworks
- Conceptual model linking variables

## 2.4 Review of Existing Systems, Models, and Approaches

- Global studies
- Regional/African studies
- Technology-based approaches

## 2.5 Case Studies Review

- Description of selected case studies
- Context, methodology, tools, results
- Strengths and weaknesses

## 2.6 Critical Analysis of the Literature

- Comparative discussion
- Methodological gaps
- Technological gaps
- Contextual limitations

## 2.7 Summary Table of Literature Review

**Author Year Context Method Tools Key Findings Limitations**

## 2.8 Research Gap Refinement

- Synthesis of gaps
- Justification of proposed system or model

## 2.9 Chapter Summary

# CHAPTER 3: TOOLS AND METHODOLOGY FOR SYSTEM DEVELOPMENT

## 3.1 Chapter introduction / Purpose of the Chapter

To explain **how the research and system were designed and developed**.

## 3.1 Chapter introduction

## 3.2 Research Methodology

- Research type (applied, experimental, design science)
- Research approach (qualitative, quantitative, mixed)
- Justification of choice

## 3.3 Data Collection Methods

- Surveys, interviews, sensors, logs
- Sampling techniques
- Instruments used

## 3.4 Model formulation and design principle (very important)

- Model origin and justification
- Conceptual diagram
- Principles and assumptions
- Equations or formal logic (if any)
- Model limitations

***3.4 Adopted Model and Underlying Principles*** (Alternative subtitle of 3.4)

**If you are adopting an existing model as-is**

**Use when:**

- You use a known model/framework without modification
- Example: TAM, UTAUT, WHO model, WCAG framework

**What to include:**

- Origin of the model
- Core principles
- Why it fits your problem
- Limitations of the original model

***3.4 Proposed Modified Model and Design Principles*** (Alternative subtitle of 3.4)

**This is the most valued option for Master theses**

**What to include:**

- Original model (brief recall)
- Identified weaknesses
- Your modifications (clearly justified)
- New parameters or components
- Diagram of original vs modified model

***3.4 Proposed Model Design and Mathematical/Formal Foundations*** (Alternative subtitle of 3.4)

**If you are designing your own model**

**Use when:**

- You introduce a new workflow, framework, or logic
- You define your own system logic

**What to include:**

- Design assumptions
- Logical flow
- Theoretical justification
- Model diagram
- Formal description

**3.4 Model Formulation and Governing Equations** (Alternative subtitle of 3.4)

**If equations or formal logic are central (AI, optimization, health metrics)**

**What to include:**

- Variable definitions
- Equations
- Constraints
- Assumptions
- Interpretation (VERY IMPORTANT)

**Clarification**

- **Bachelor thesis**: Conceptual model + diagram may be enough
- **Master thesis**: Model + justification + (optional) equations
- **PhD**: Model + formal proof + validation

### 3.5 Parameter Identification and Specification

- Functional parameters
- Technical parameters
- User-related parameters

### 3.6 System Development Methodology

- SDLC model (Agile, Waterfall, Design Science)
- Development phases

### 3.7 System Architecture

- High-level architecture
- Hardware/software architecture
- Network or cloud architecture
- Flowchart
- Workflow diagram
- Flow diagram
- Logigram
- Organogram (if institutional)

### 3.8 Tools and Technologies Requirements

- Hardware requirements
- Software requirements
- Programming frameworks
- Development platforms

### 3.9 System Components and Modules

- Description of each component
- Interaction between components

### 3.10 System Functional Requirements

- User requirements
- System requirements
- Non-functional requirements

## 3.11 Evaluation Metrics Definition

- Performance metrics
- Usability metrics
- Accessibility metrics
- Reliability and scalability metrics

## 3.12 Chapter Summary

# CHAPTER 4: IMPLEMENTATION AND SYSTEM EVALUATION

## 4.1 Chapter introduction / Purpose of the Chapter

To present **implementation details, testing, evaluation, and comparison**.

## 4.2 Implementation Environment

- Hardware environment
- Software environment
- Development settings

## 4.3 System Prototyping and Implementation

- Interface design
- Module implementation
- Screenshots or diagrams

## 4.4 System Functional Testing

- Unit testing
- Integration testing
- User acceptance testing

## 4.5 System Performance and Metrics Evaluation

- Measurement results
- Data analysis
- Graphs and tables

## 4.6 User Evaluation and Feedback (if applicable)

- Participants
- Evaluation protocol
- Results

## 4.7 Comparative Analysis with Existing Systems

- Benchmarking
- Comparison with recent literature

## 4.8 Discussion of Results

- Interpretation of findings
- Alignment with objectives

## 4.8 Chapter Summary

# CHAPTER 5: CONCLUSION AND PERSPECTIVES

## 5.1 Chapter introduction / Purpose of the Chapter

To synthesize contributions, reflect critically, and project future work.

## 5.2 Summary of the Research Work

- Recall of objectives and approach
- Evolution of the work

## 5.3 Discussion of Results

- Scientific implications
- Technical implications
- Societal implications

## 5.4 Contributions of the Study

- Academic contribution
- Technical/system contribution
- Practical contribution

## 5.5 Limitations and Challenges Encountered

- Technical
- Methodological
- Environmental

## 5.6 Perspectives and Future Work

- System enhancement
- Research extension
- New application domains

## 5.7 Monetization and Sustainability Plan (if applicable)

- Business model
- Deployment strategy
- Stakeholder engagement

## 5.8 Applicability and Utility of the System

- Real-world deployment potential
- Beneficiaries

### 5.9 General Conclusion

# FINAL SECTIONS

- **References** (APA / IEEE – consistent style)
    - Books/thesis
    - Journals
    - Internet pages/documents
- **Appendices** (questionnaires, code, extra results)

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